Markets

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Chapter 3: Supply and Demand

Learning Objectives

1. What is a competitive market?
2. What is demand?, how does it change, and what factors determine it?
3. What is supply, how does it change, and what factors determine it?
4. How do supply and demand jointly determine market equilibrium?
5. Why does the market tend to move toward equilibrium?
6. How does market equilibrium respond to changes in underlying economic conditions?

Competitive Markets

- A market is a place where goods or services are bought and sold.

- A competitive market is one with **many buyers and sellers of the same good** (or service), none of whom can individually influence the price.

- The supply and demand model describes how a competitive market functions.

Supply and Demand Model (In a Nutshell)

- Buyers determine demand.
- Sellers (or producers) determine supply.
- Interaction between demand and supply determines market outcome.
  - In particular, price and quantity.

Demand

Quantity Demanded

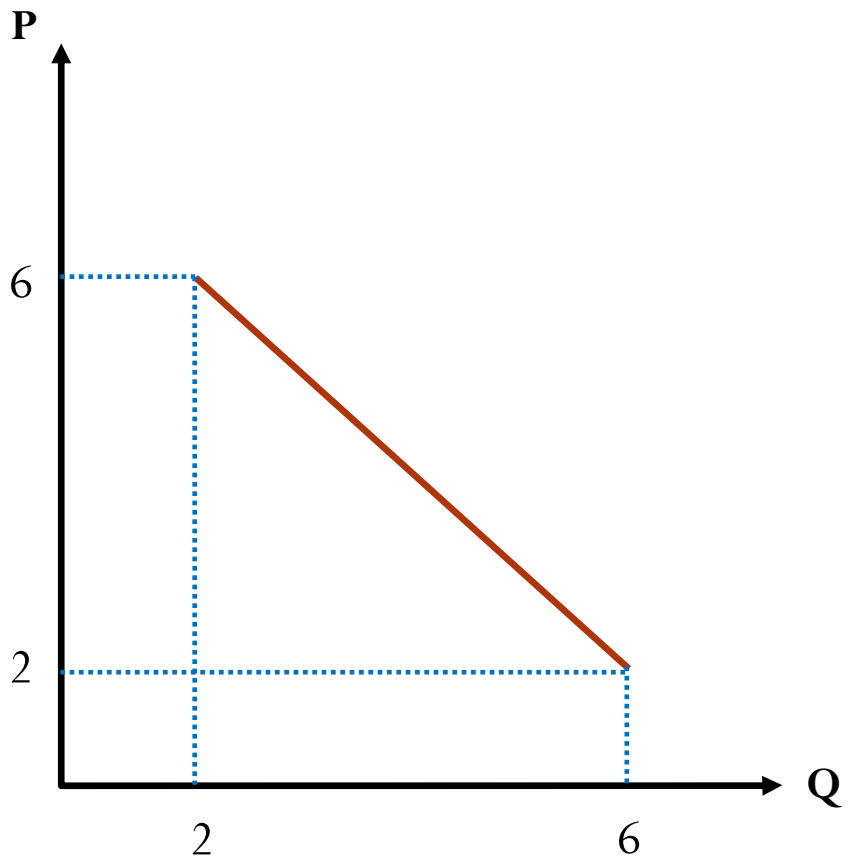
The amount of a good (or service) that buyers are willing and able to purchase at a given price.

Demand

A schedule showing the quantity of a good that consumers wish to purchase at each price.

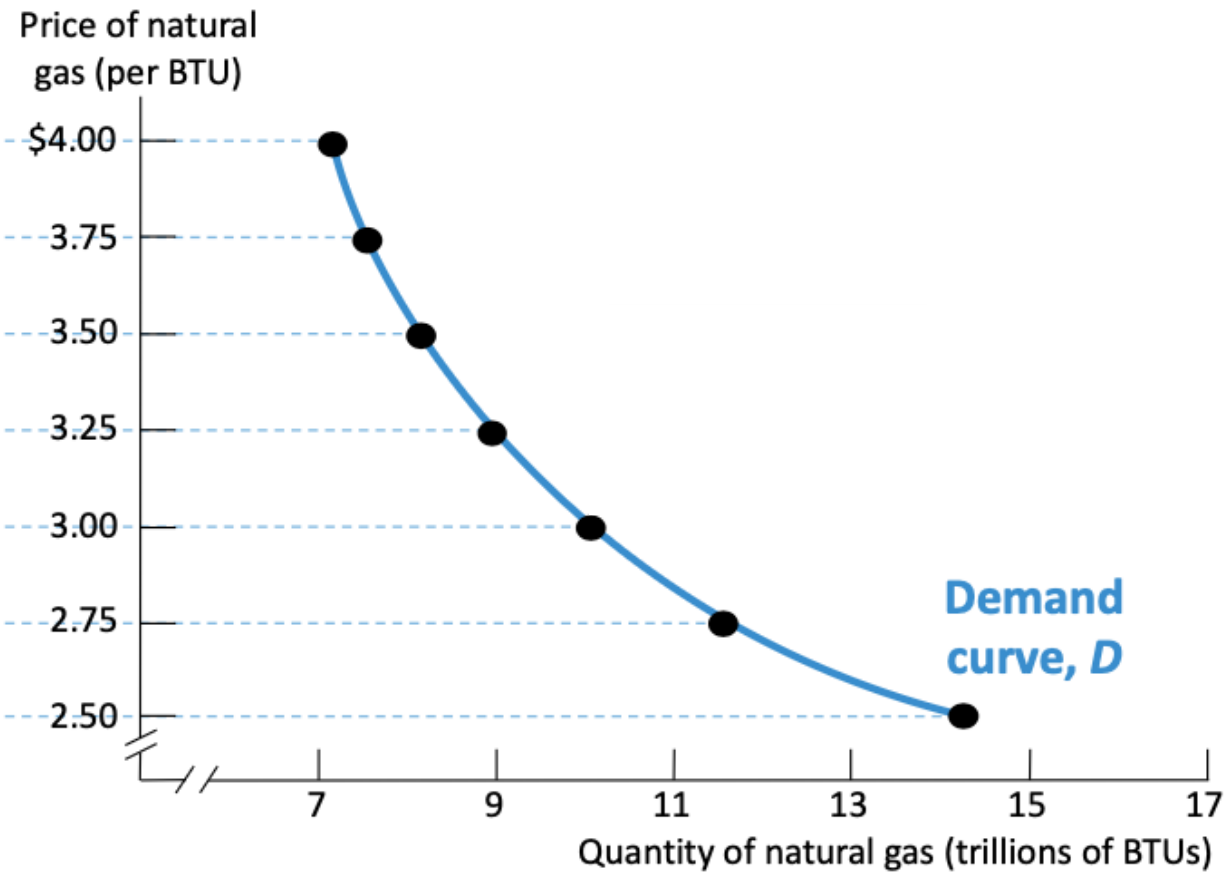
Demand Schedule → Demand Curve

Price per Slice of Pizza	# of Slices Demanded
2	6
3	4
4	3
5	2.4
6	2

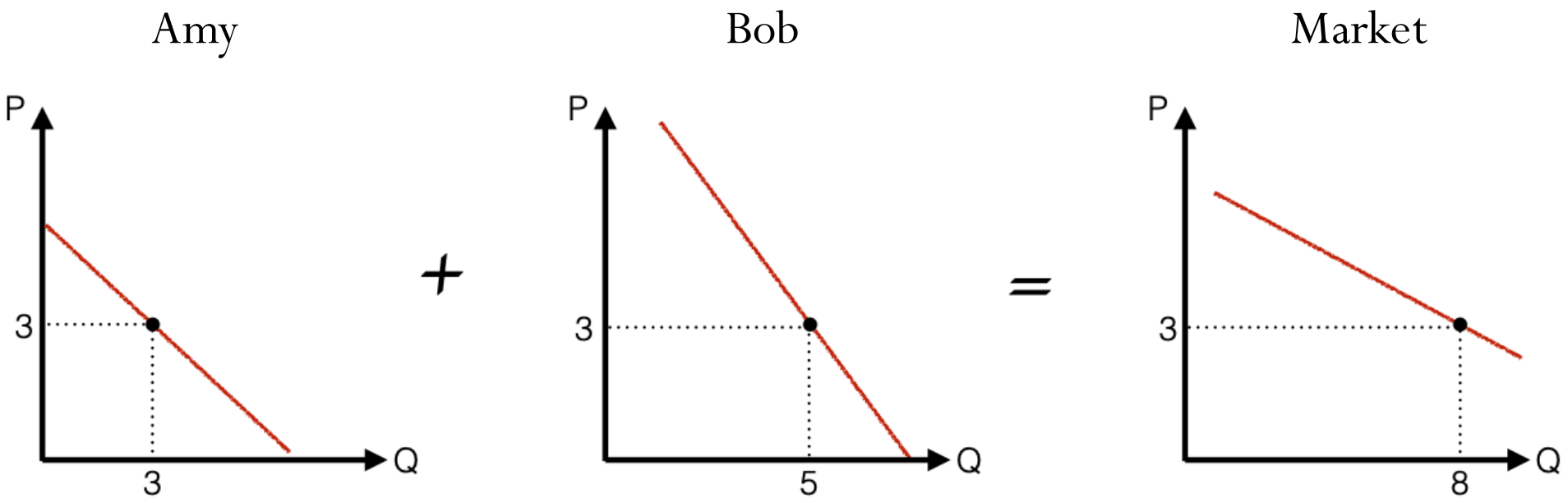


Demand Schedule → Demand Curve

Price of natural gas (per BTU)	Quantity of natural gas demanded (trillions of BTUs)
$4.00	7.1
$3.75	7.5
$3.50	8.1
$3.25	8.9
$3.00	10.0
$2.75	11.5
$2.50	14.2



Individual Demand → Market Demand



The market demand (curve) is the sum of individual demand (curves)

The Law of Demand

Other things equal, the quantity demanded of a good falls when the price of the good rise.

The Law of Demand

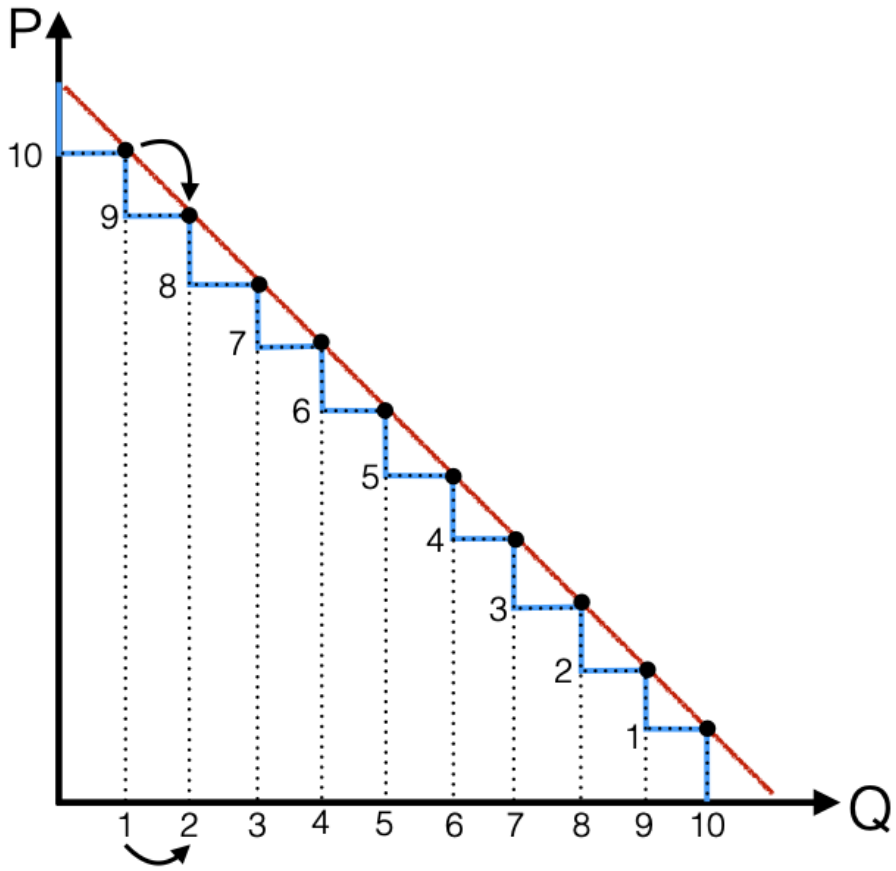
- If each individual demand satisfies the law of demand, the aggregate market demand will also satisfy it.

- The law can hold at the market level even when individual demand curves are not "smooth".

- For instance, suppose that each consumer has unit demand and a maximum willingness to pay (WTP).

Marginal Buyers

- Let there be 10 potential buyers with WTP of 1,2,…,10.
- As price decreases, more buyers enter the market.
- Buyers with higher WTP enter first.
- The marginal (or last) buyer defines the market demand at that price.

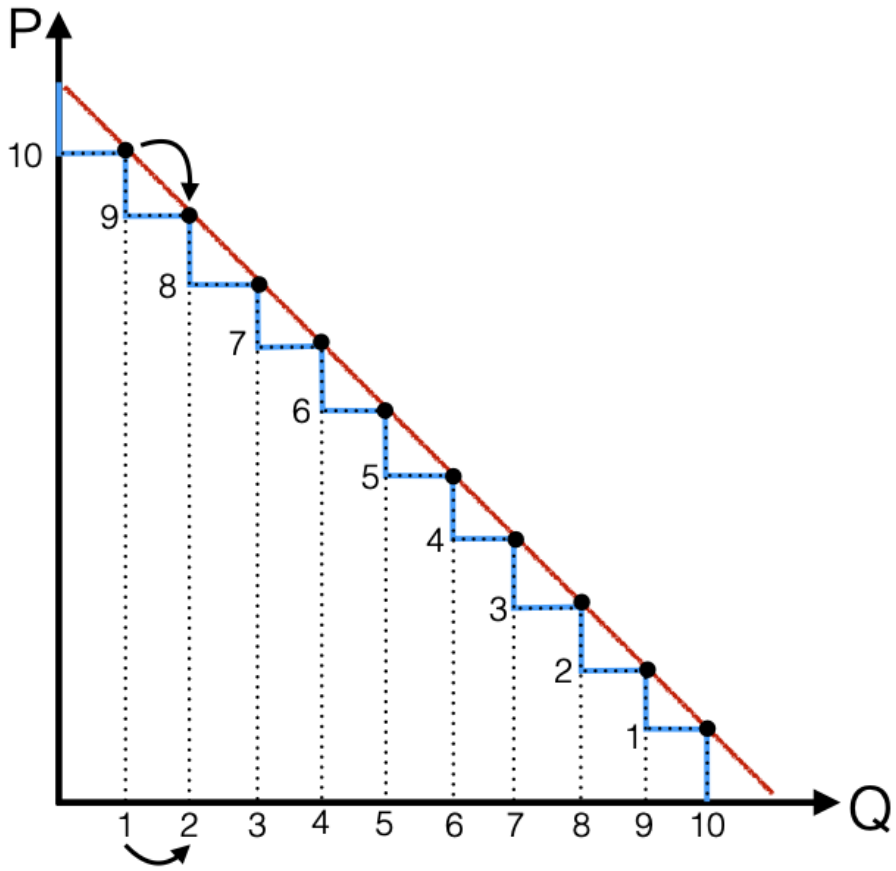


Marginal Benefit Curve

The marginal benefit curve represents the change in total benefit to society from producing one additional unit of good.

Demand (Curve) = Marginal Benefit (Curve)

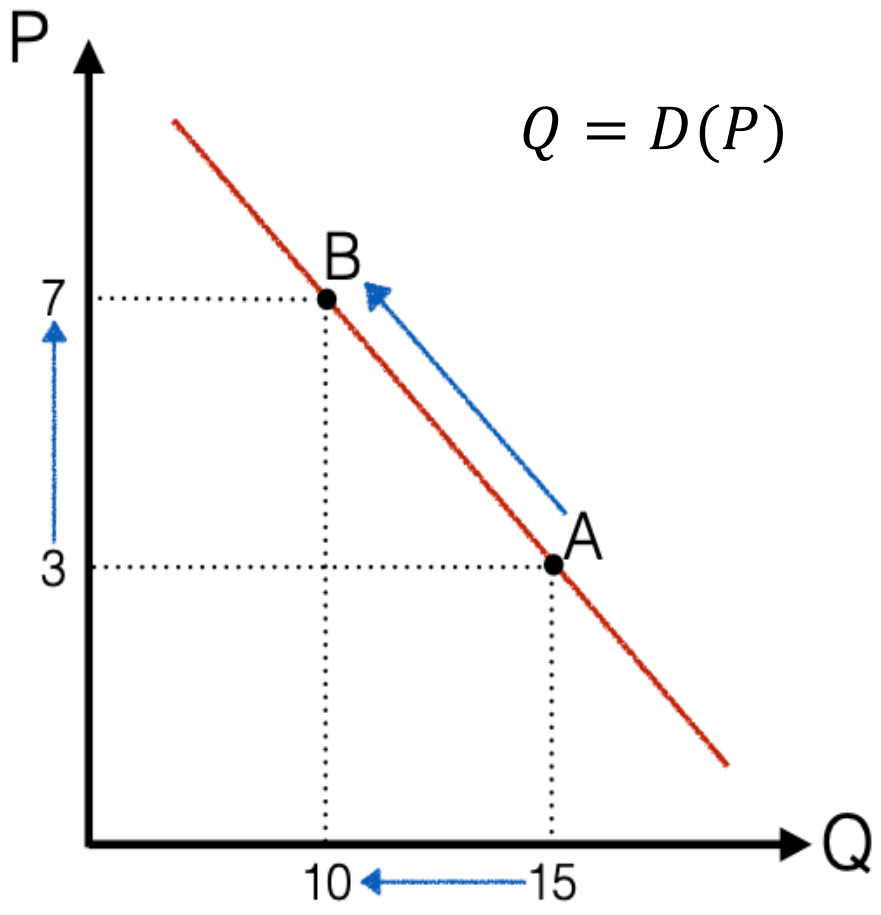
Buyer	WTP	TB	MB
1st	10	10	10
2nd	9	19	9
3rd	8	27	8
4th	7	34	7
5th	6	40	6
6th	5	45	5
7th	4	49	4
8th	3	52	3
9th	2	54	2
10th	1	55	1



Change in Quantity Demanded

- Change in quantity demanded is caused by a change in the price of the product.

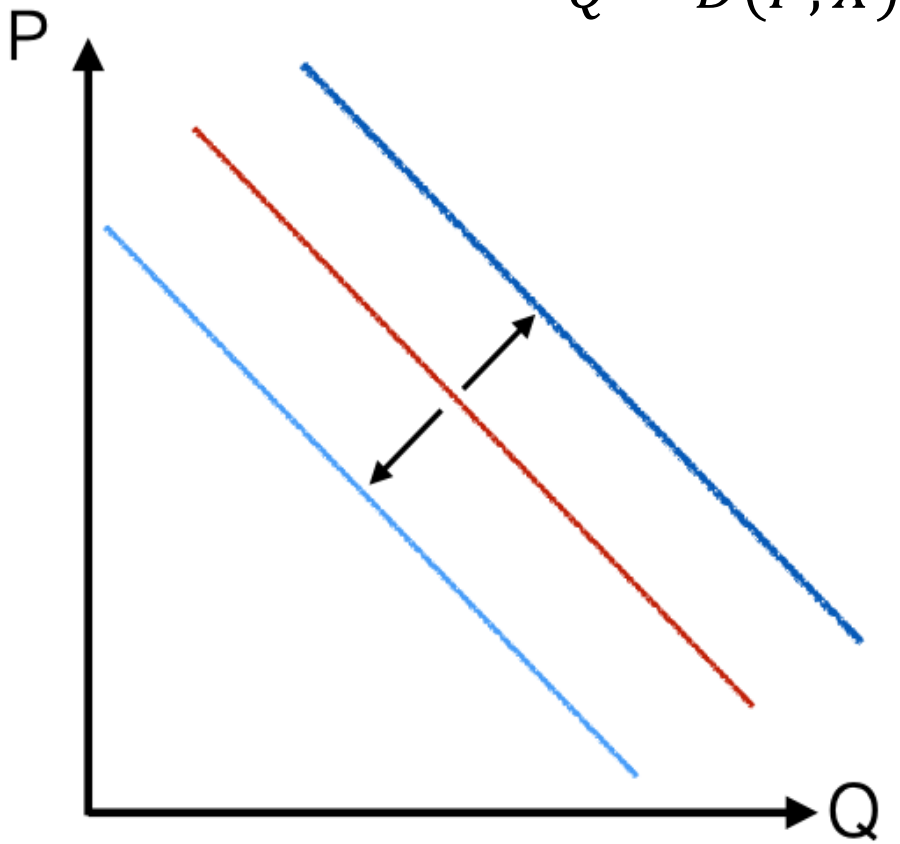
- A movement along the demand curve.



Change in Demand

- Change in demand is caused by a change in determinant other than the price.
- A shift in the demand curve:
  - outward shift = increase in demand
  - inward shift = decrease in demand

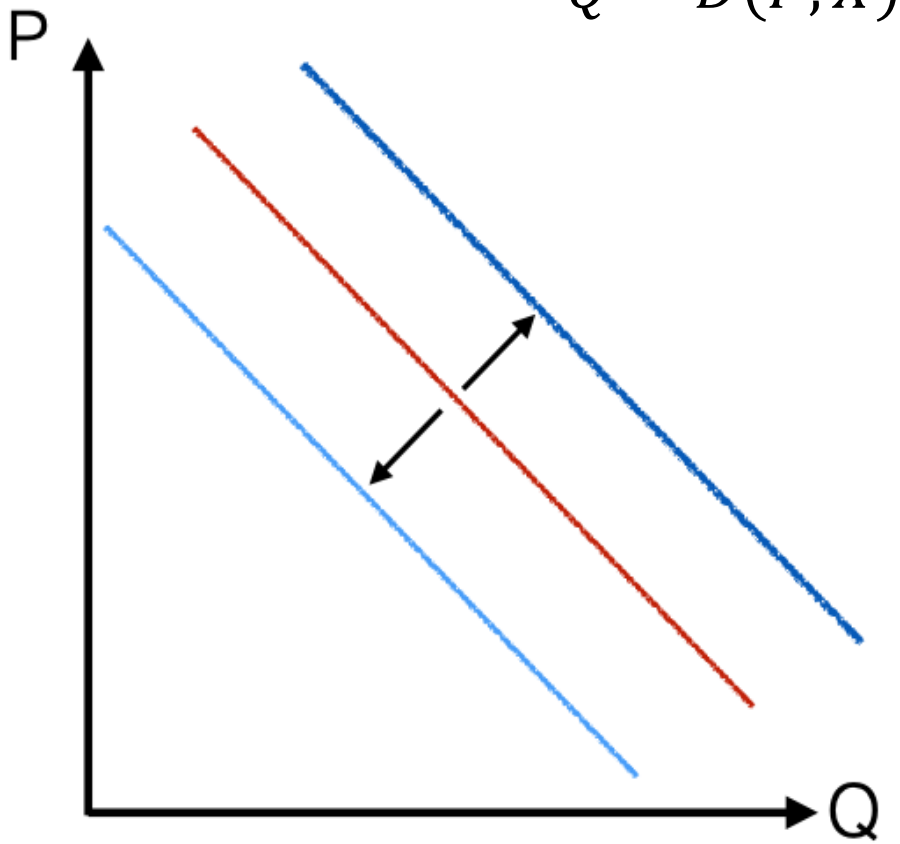
$$Q = D(P, X)$$



What is in X?

- Taste or preferences
- Expectation
- Number of consumers
- Income
- Price of related goods or services
- ……

$$Q = D(P, X)$$



Consumer Income

- The effect of income changes on demand depends on the nature of the good.
- If demand rises with income, the product is a **normal good**.
- If demand falls with income, the product is an **inferior good**.

Price of Related Goods

- Two goods are **substitutes** if a decrease in the price of one good leads to a decrease in the demand of the other good.
  - Example: iPhone vs. Android
- Two goods are **complements** if a decrease in the price of one good leads to an increase in the demand of the other good.
  - Example: Streaming services and smart TVs

Supply

Quantity Supplied

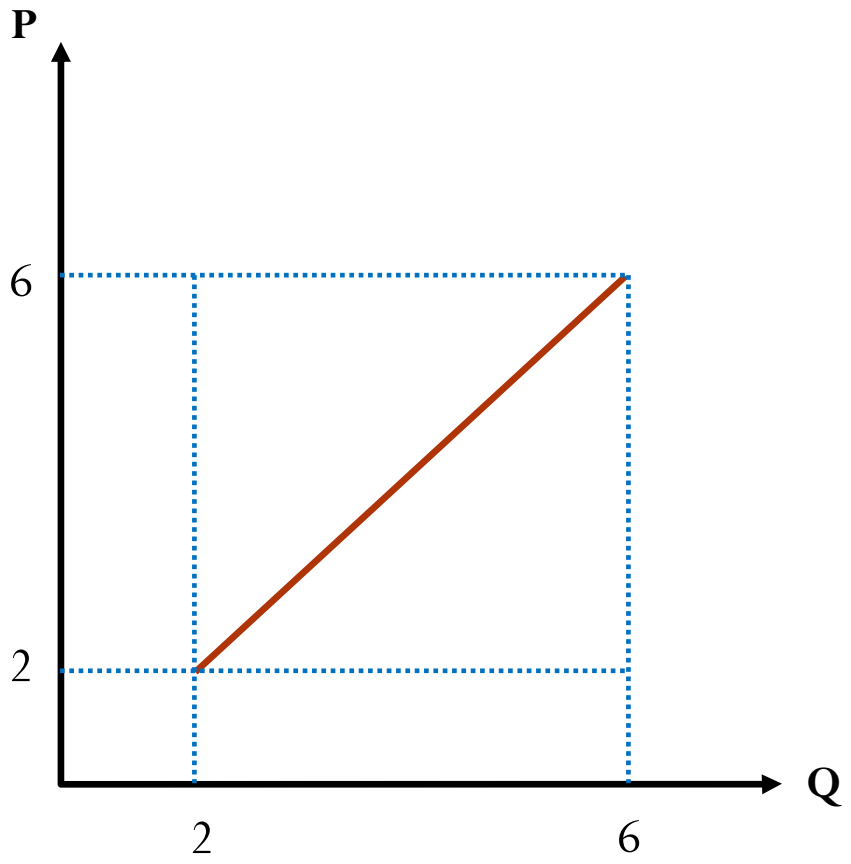
The amount of a good (or service) that sellers are willing and able to sell at a given price.

Supply

A schedule showing the quantity of a good that sellers wish to sell at each price.

Supply Schedule → Supply Curve

Price per Tutoring Hour	# of hours Supplied
2	2
3	3
4	4
5	5
6	6



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Price of natural gas (per BTU)	Quantity of natural gas supplied (trillions of BTUs)
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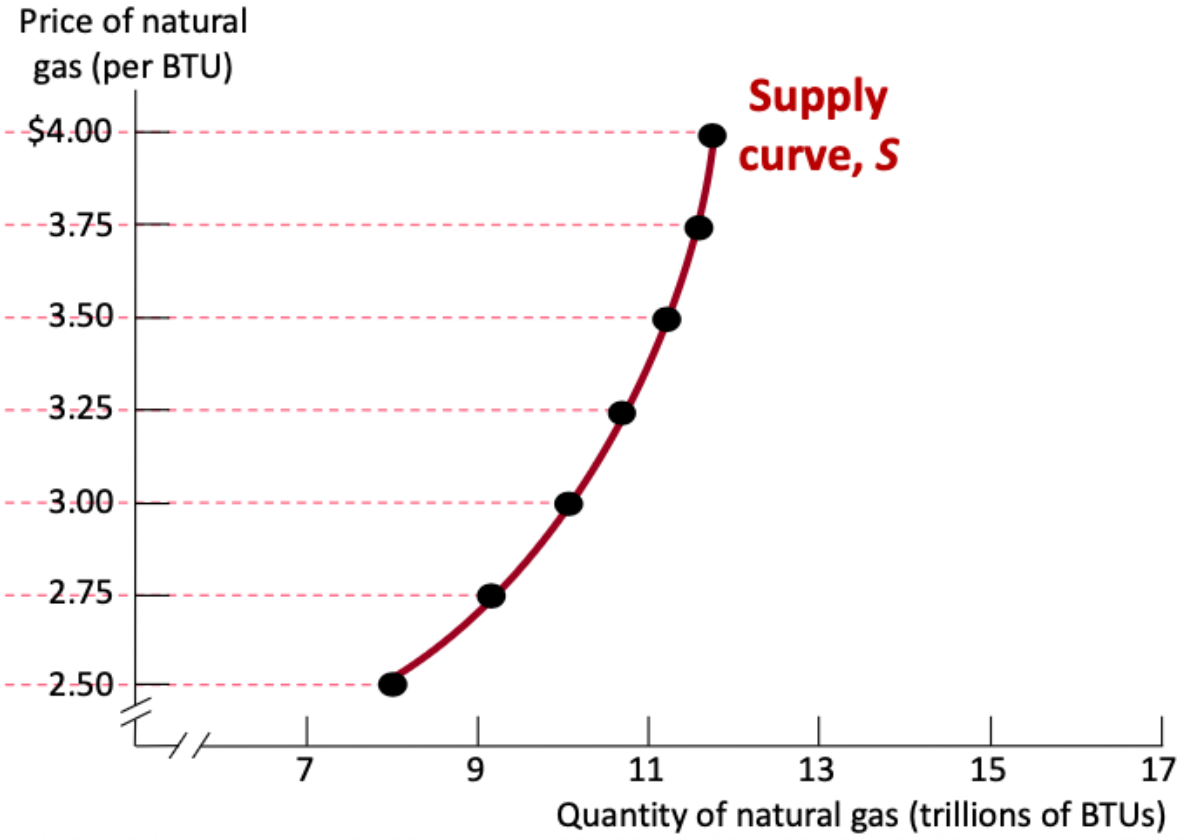
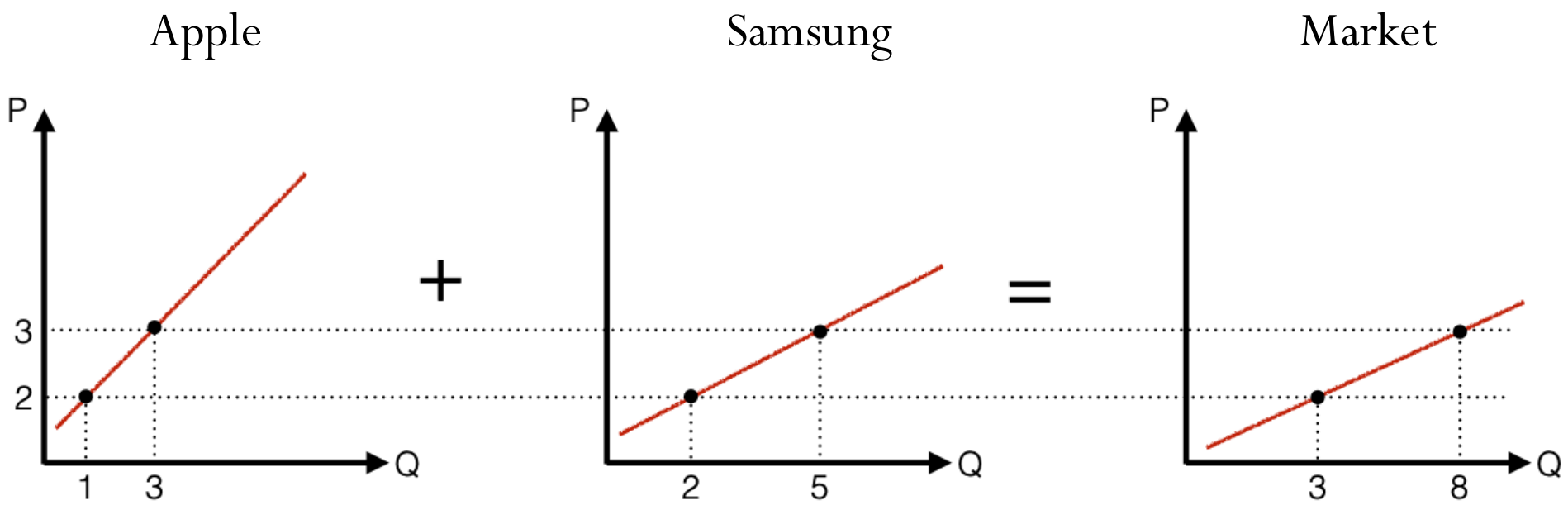


Individual Supply → Market Supply



The market supply (curve) is the sum of individual supply (curves)

The Law of Supply

Other things equal, the quantity supplied of a good rises when the price of the good rise.

The Law of Supply

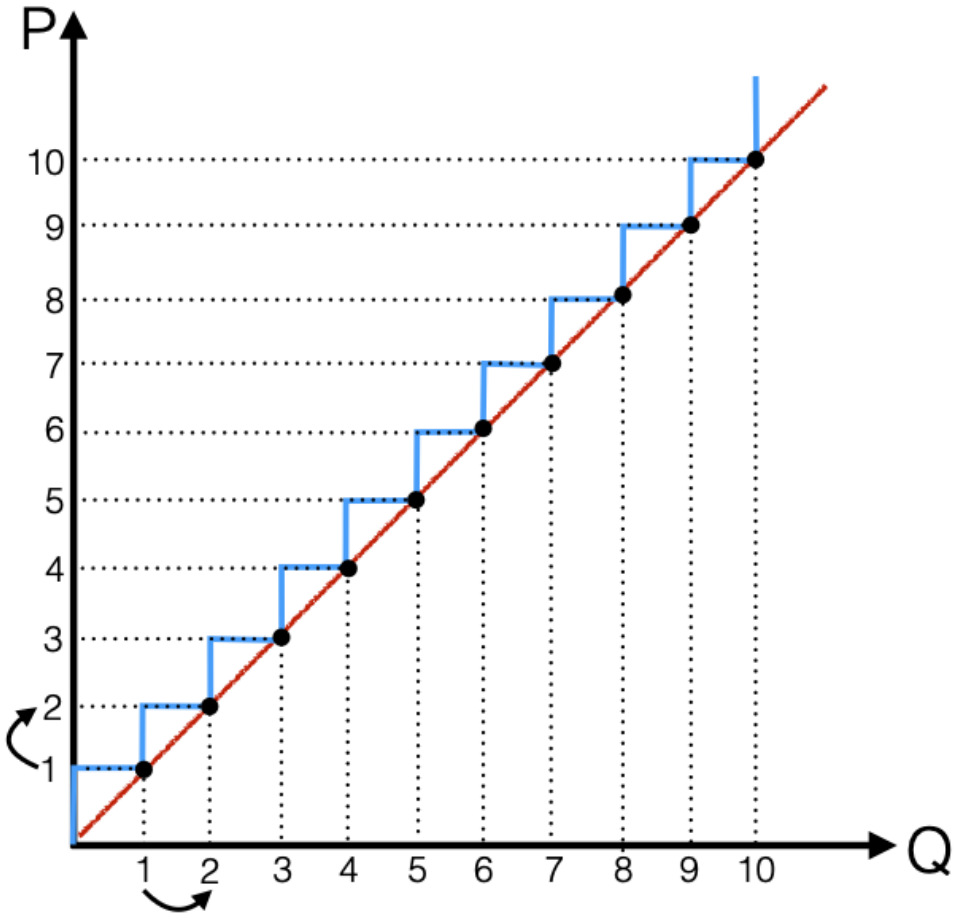
- If each seller's supply satisfies the law of supply, the aggregate market supply will also satisfy it.

- The law can hold at the market level even when individual supply curves are not "smooth".

- For instance, suppose that each seller has unit supply and a minimum willingness to accept (WTA, or production cost).

Marginal Sellers

- Let there be 10 potential sellers with WTA of 1,2,…,10.
- As price increases, more sellers enter the market.
- Sellers with lower WTA enter first.
- The marginal (or last) seller defines the market supply at that price.

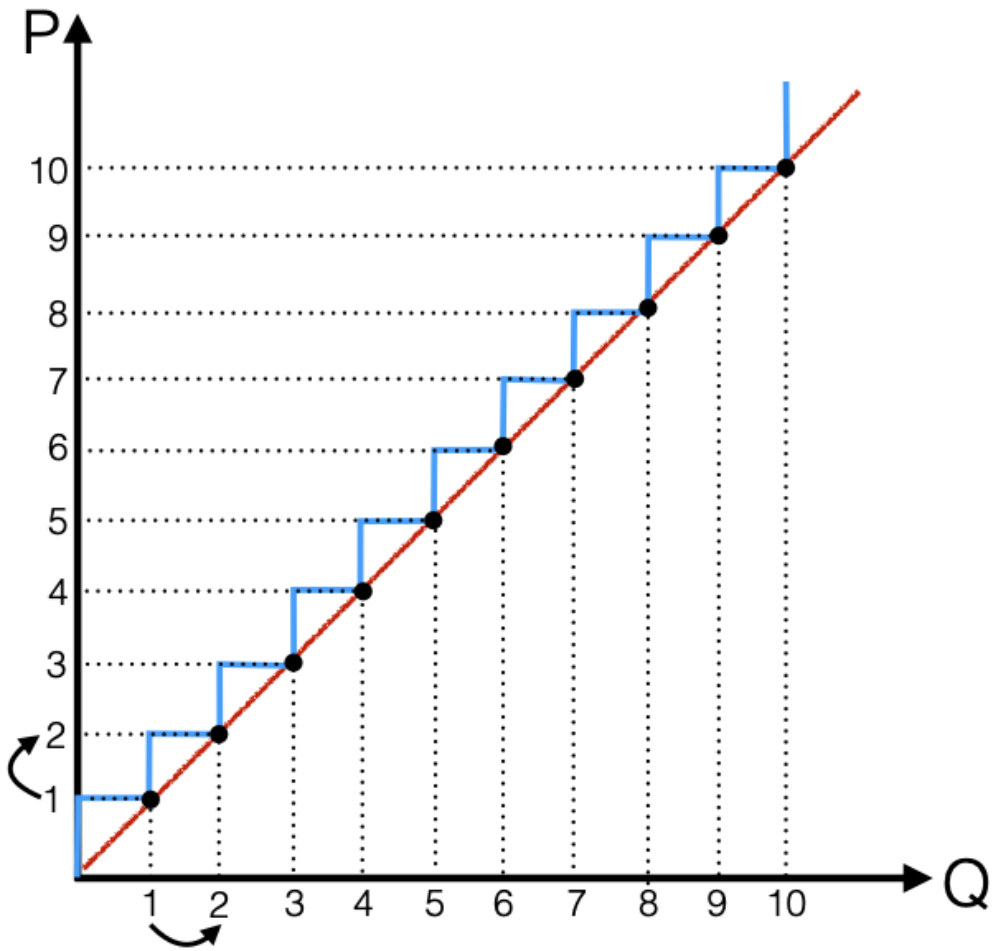


Marginal Cost Curve

The marginal cost curve represents the change in total cost to society from producing one additional unit of good.

Supply (Curve) = Marginal Cost (Curve)

Seller	Cost	TC	MC
1st	1	1	1
2nd	2	3	2
3rd	3	6	3
4th	4	10	4
5th	5	15	5
6th	6	21	6
7th	7	28	7
8th	8	36	8
9th	9	45	9
10th	10	55	10



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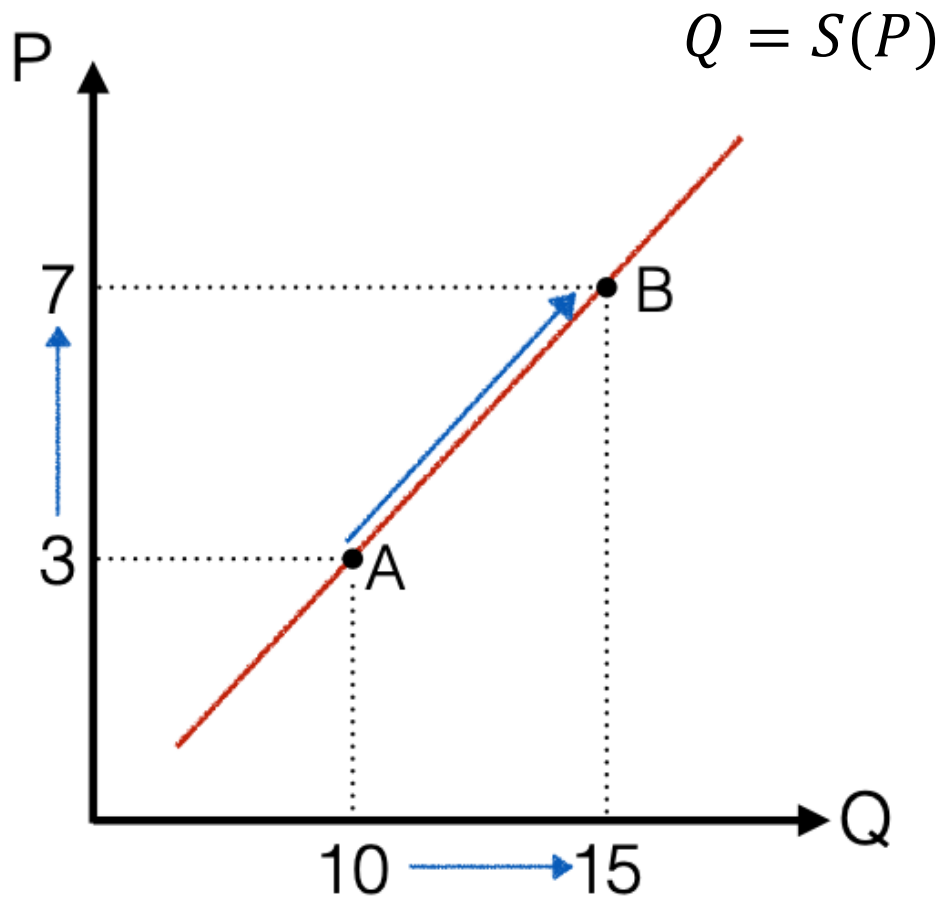

Change in Supply

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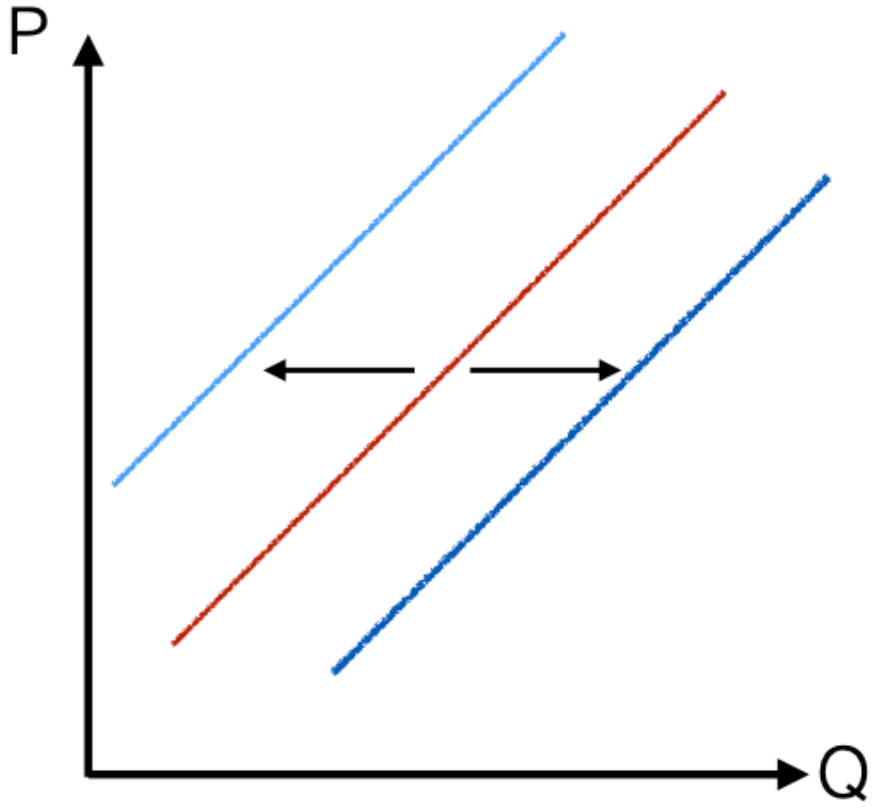
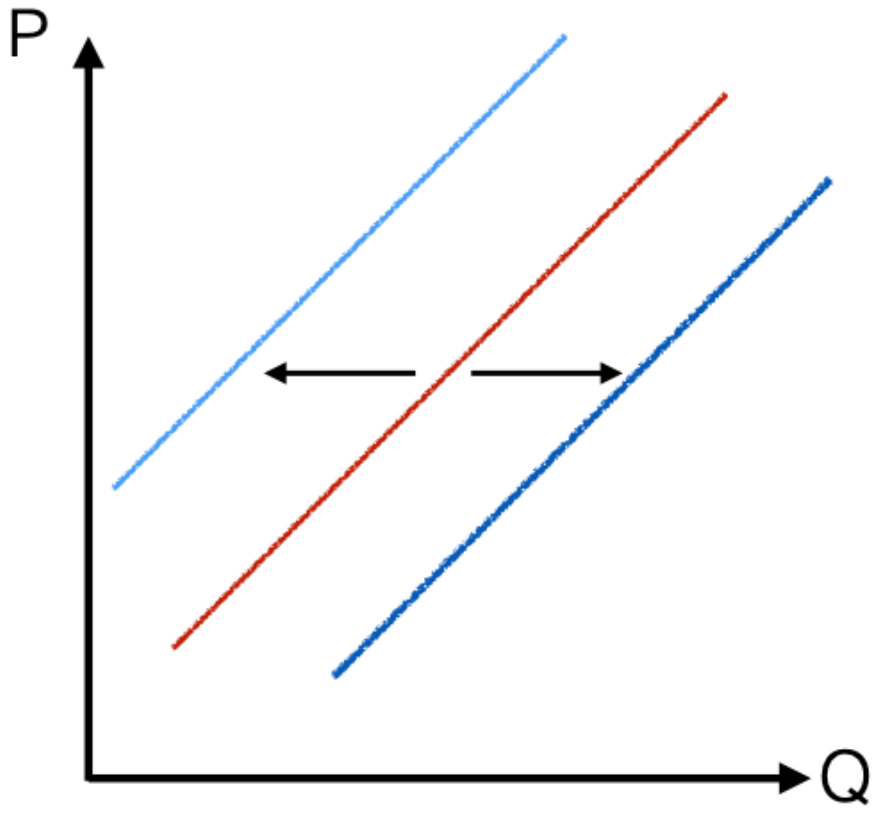


What is in X?

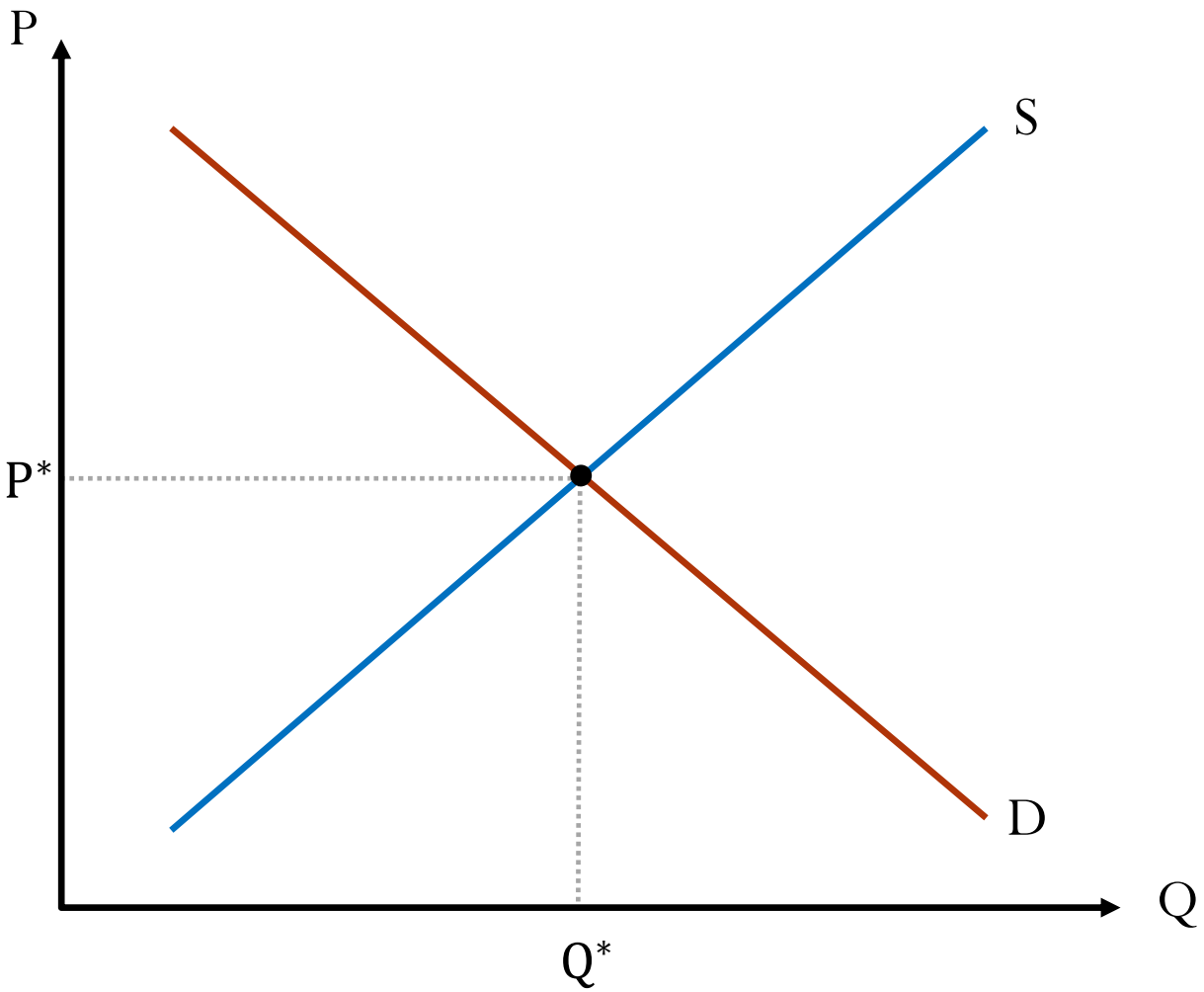
- Input prices
- Price of related goods
- Technologies
- Expectation
- Numbers of producers
- ……

$$Q = S(P, X)$$



Market Equilibrium

Demand, Supply, and Equilibrium



Equilibrium Price and Quantity

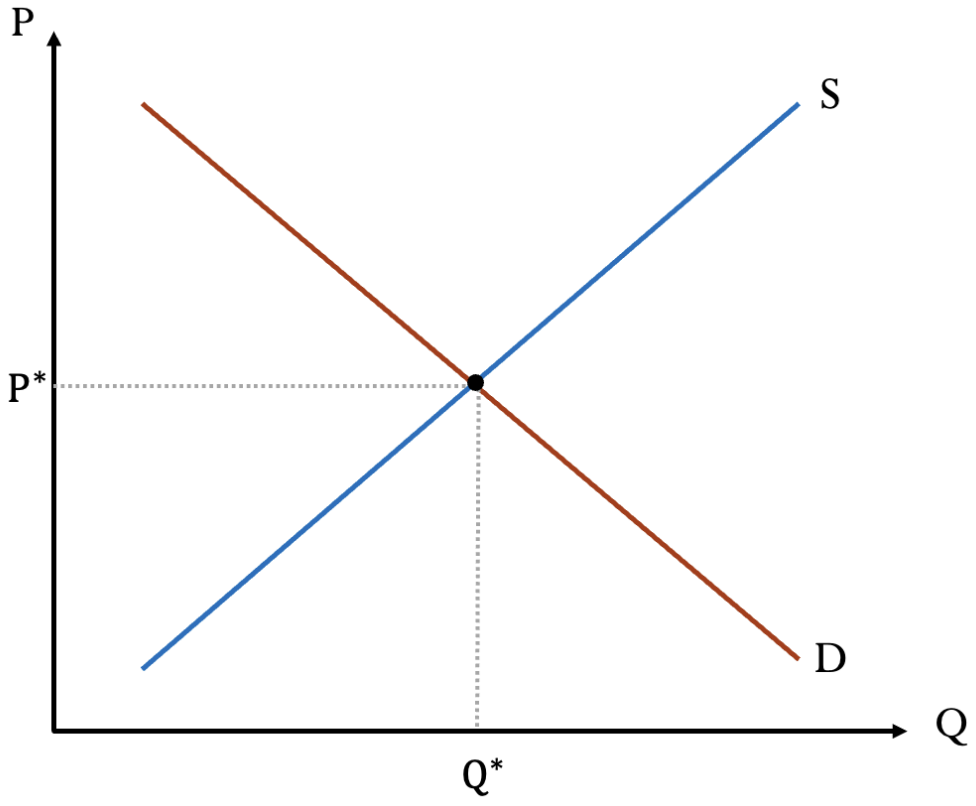
Equilibrium Price: P^*

- The price at which supply equals demand

Equilibrium Quantity: Q^*

- The quantity at which supply equals demand

Graphically, (P^*, Q^*) is the intersection of the supply and demand curves.



Market Clearing

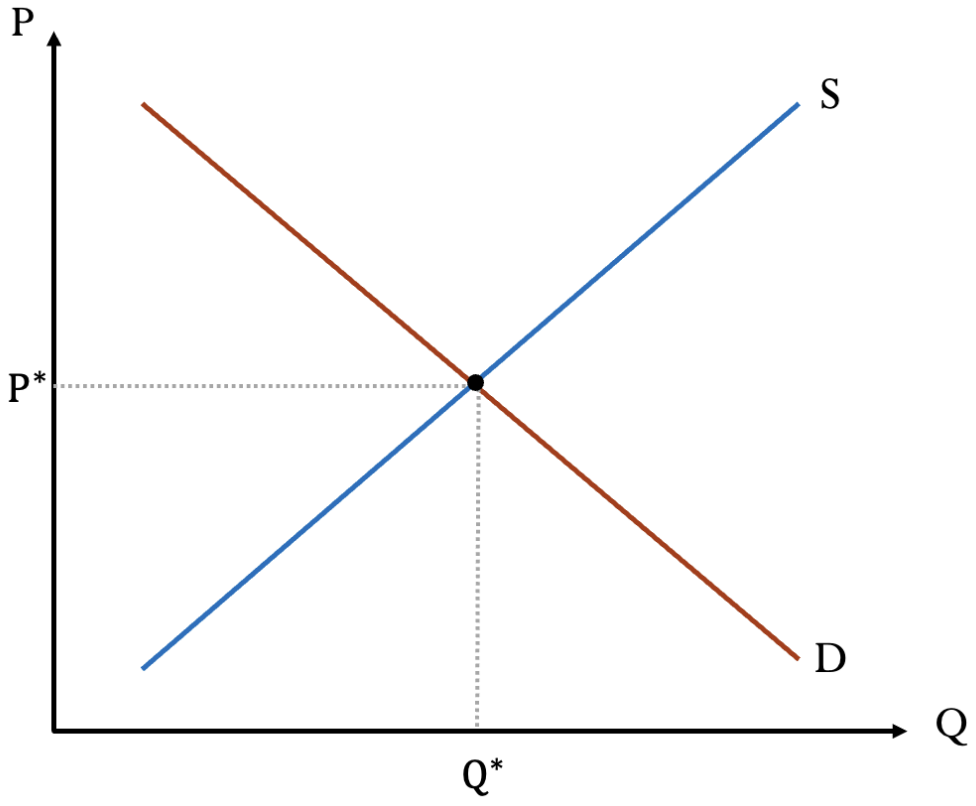
Given the equilibrium price,

Number of Buyers willing to buy

||

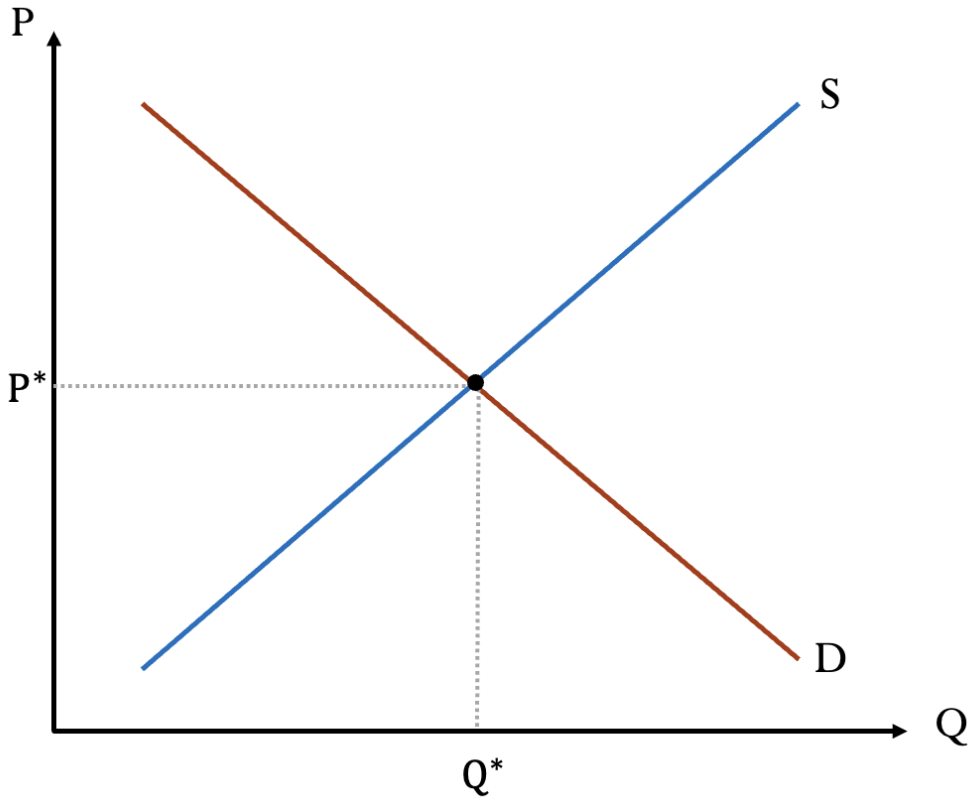
Number of Sellers willing to sell

Thus, the equilibrium price is often referred as the market-clearing price.

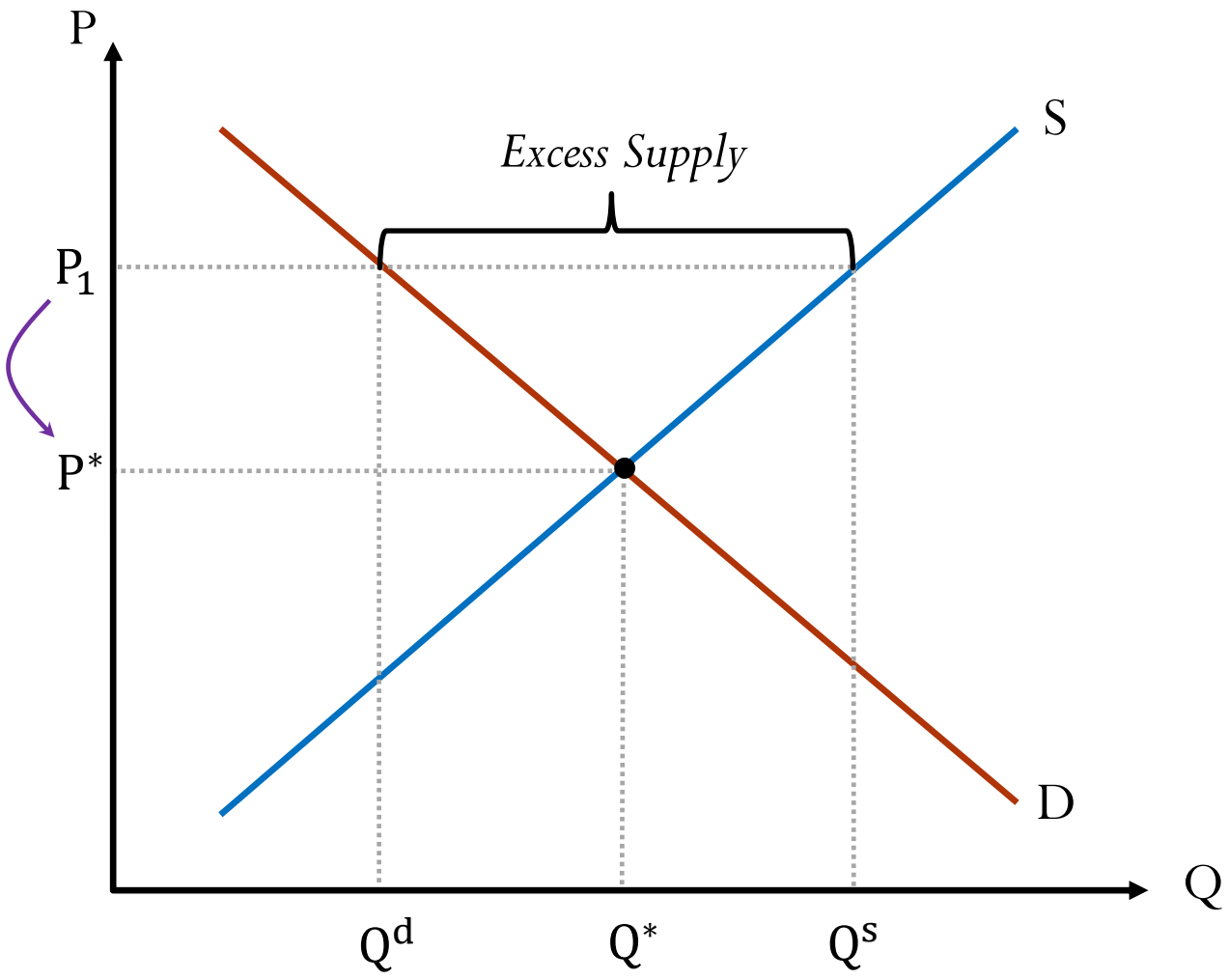


Why Does the Market Move Toward Equilibrium?

- So far, we have not explained why market outcomes converge to equilibrium.
- The key lies in understanding the forces that drive market adjustment.
- That is, how the market responses when outcome deviates from equilibrium.

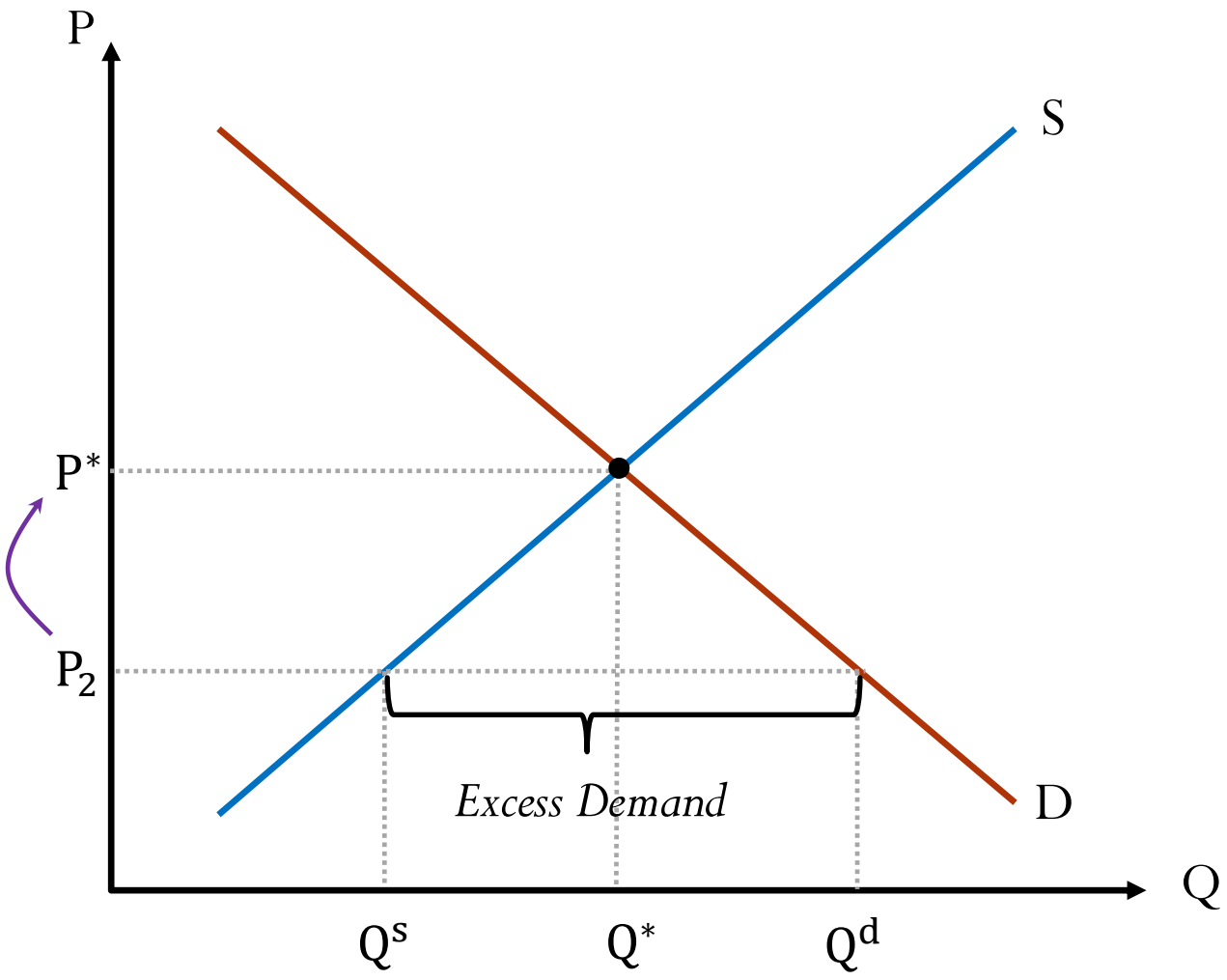


The Price Mechanism



- Suppose the current price is $P_1 > P^*$
- At this price, the market demand is Q^d, while the market supply is Q^s.
- $Q^s > Q^d \Longrightarrow$ Excess supply in the market.
- Sellers unable to sell their products will cut prices to attract buyers.
- Market price tends to ↓ under this pressure.

The Price Mechanism

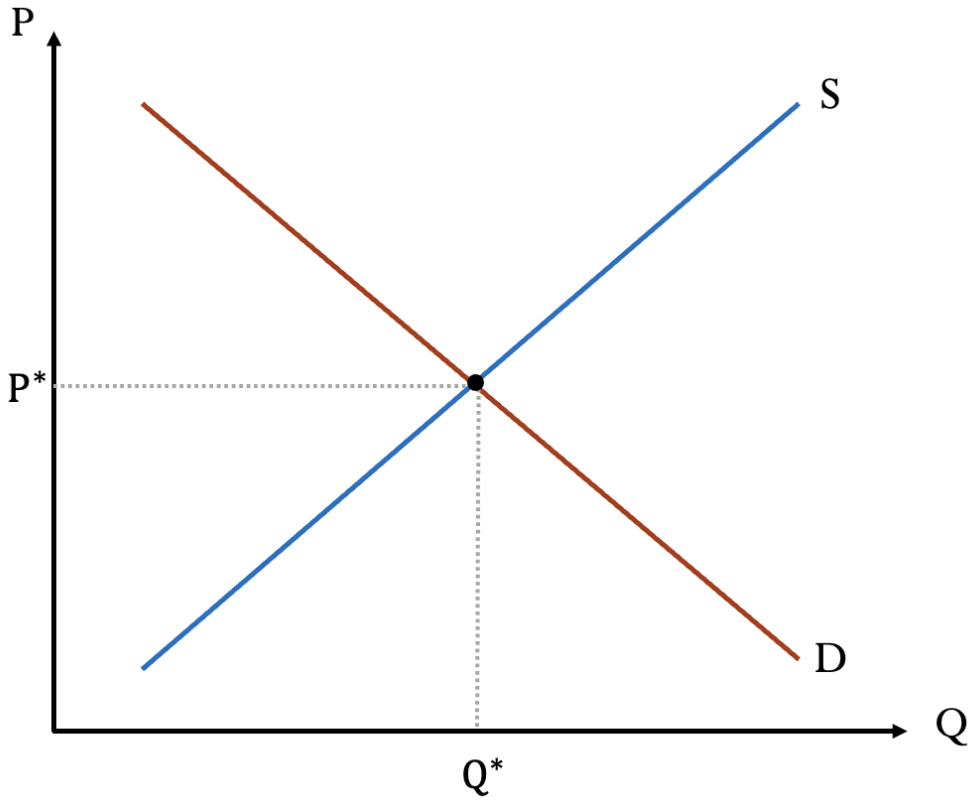


- Suppose the current price is $P_2 < P^*$
- At this price, the market demand is Q^d, while the market supply is Q^s.
- $Q^s < Q^d \Longrightarrow$ Excess demand in the market
- Buyers unable to purchase the product will compete by offering higher prices.
- Market price tends to ↑ under this pressure.

The Price Mechanism

The Law of Demand and Supply

In a free market, the price of any good adjusts to equalize the quantity supplied and the quantity demanded for that good.



Behind the Price Mechanism

- Sufficient competition among buyers and sellers ensures the price mechanism works properly.
- Important assumption behind the model:
  - No single player has market power
- Both buyers and sellers are price takers:
  - There are many buyers in the markets
  - There are many sellers in the markets
- The goods offered for sale are homogeneous.

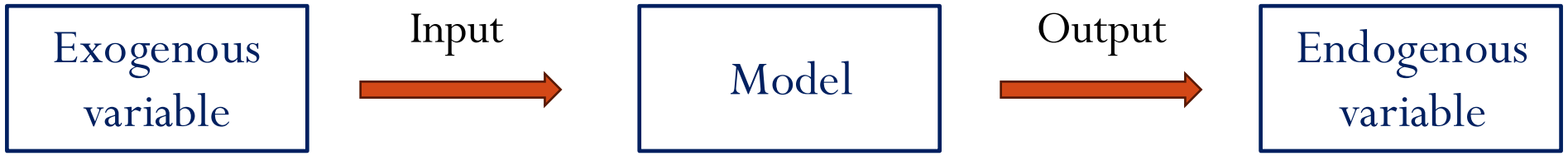
Limitations of Static Equilibrium Analysis

- The supply-demand model describes a market's long-run, static equilibrium.
- But in the real world, prices are constantly in motion and rarely sit still.
- How can we model this reality?
  - Approach A: Build a complex, dynamic model from the ground up.
  - Approach B: Extend the current model to understand dynamics through statics.

Comparative Statics

Comparative Statics

Comparative statics is the determination of the changes in the endogenous variables of a model that will result from a change in the exogenous variable or parameters of that model.



Exogenous vs. Endogenous Variables

Economic Model	Research Question	Exogenous Variable (Input)	Endogenous Variable (Output)
Demand Model	Consumer behavior	**Price**, income, price of related goods, etc.	Quantity demanded
Supply Model	Producer behavior	**Price**, input cost, technology, etc.	Quantity supplied
Supply-Demand Model	Market behavior	Consumer: income, price of related goods, etc. Producer: input cost, technology, etc.	Equilibrium **price** Equilibrium quantity

Comparative Statics: Step-by-Step

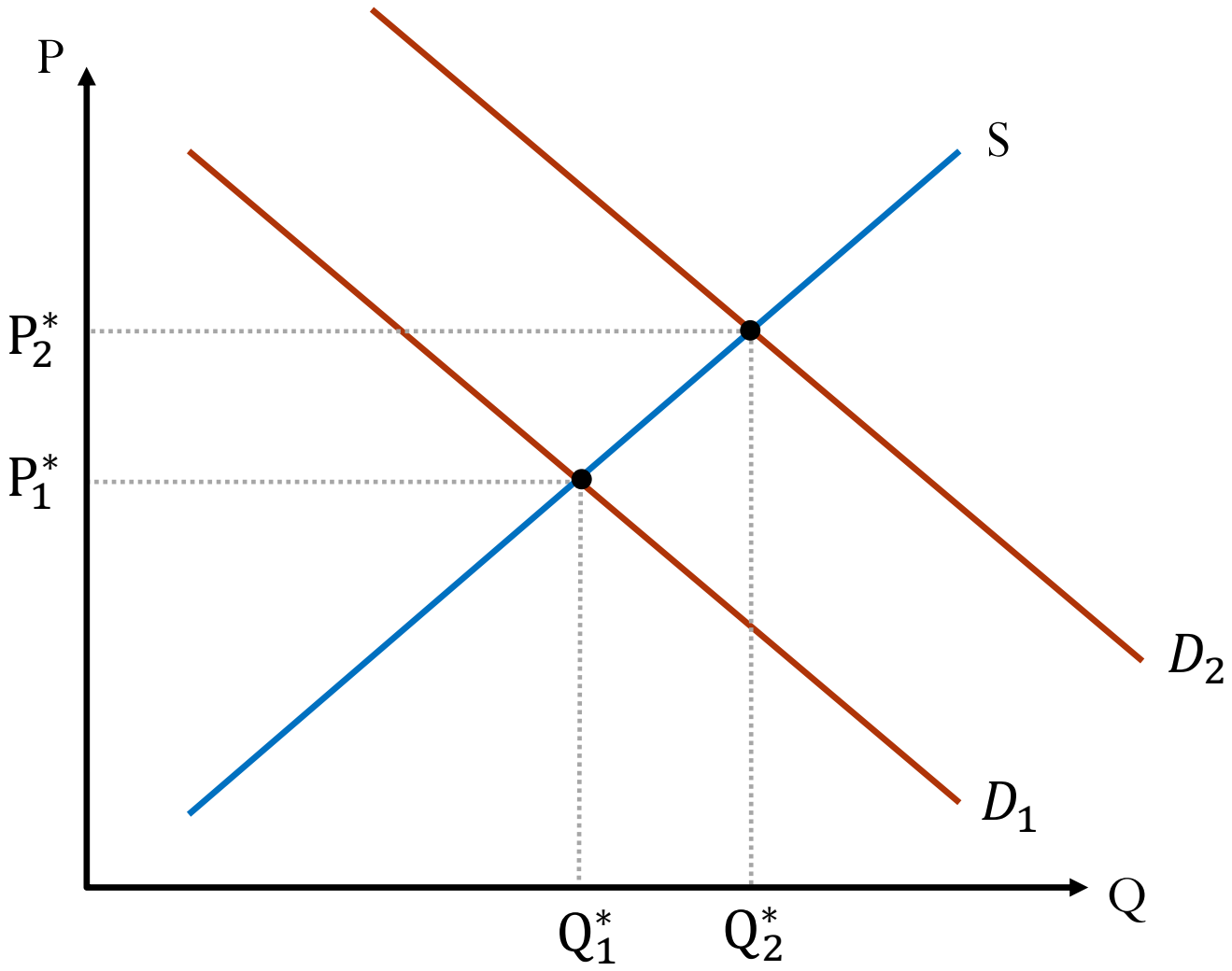
- Comparative statics analysis is a powerful tool for understanding the co-movement of price and quantity.
- It involves three simple steps:

① Identify which curve (or curves) shift.

② Identify how the curve shifts (direction).

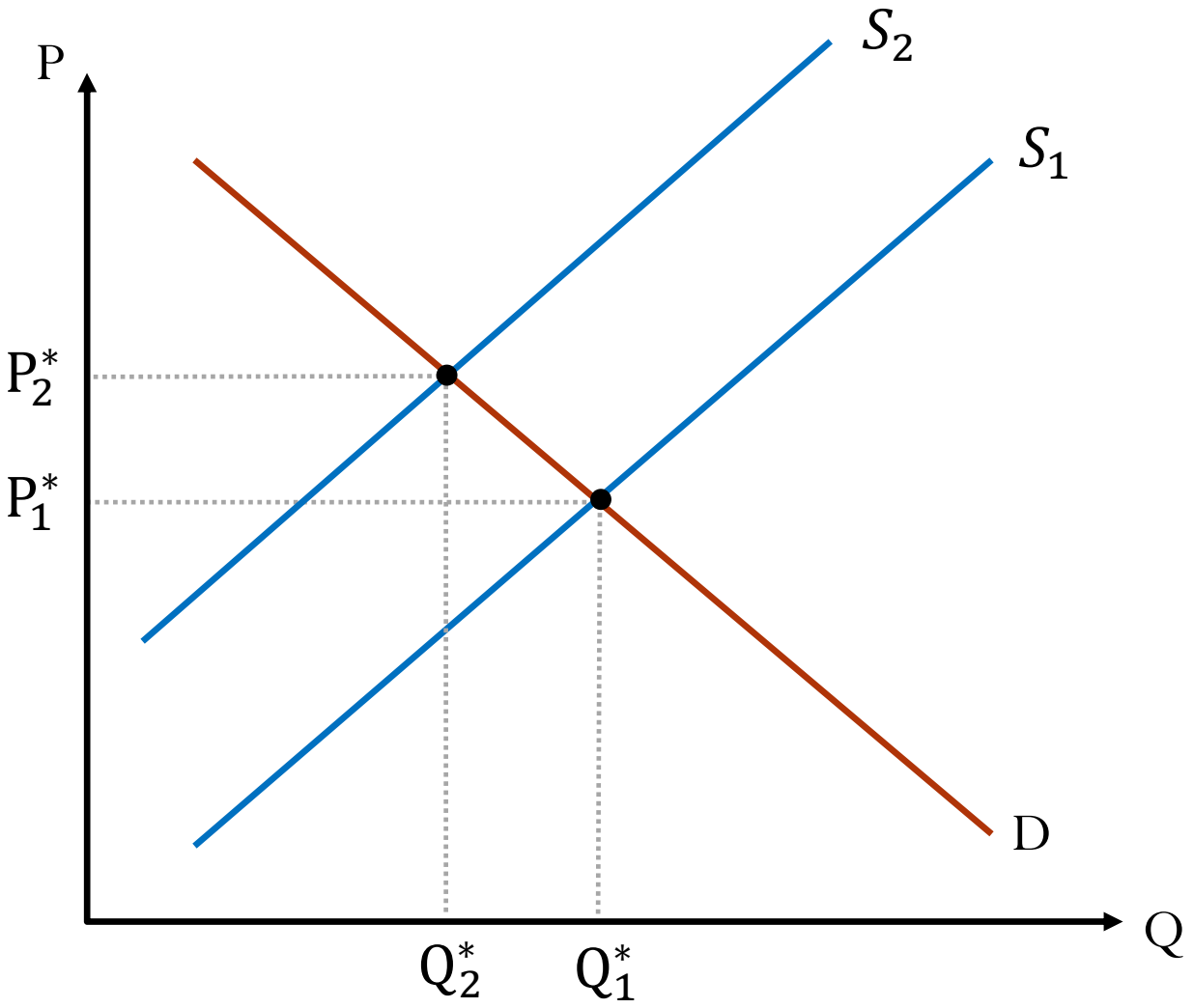
③ Locate the new equilibrium and compare it to the old one.

Warm-Up: The Market for Ice-Cream



Suppose hot weather increases the demand for ice cream.

Warm-Up: The Market for Ice-Cream



Imagine a shortage of milk reducing the supply for ice cream.

Applications

I. The housing prices are rising, but the quantity of homes sold is also increasing.

- This seems puzzling because it appears to violate the law of demand. (Does it?)

II. A government subsidies farmers on seeds, increasing the food supply.

$\Longrightarrow$ More food supply leads to lower food prices.

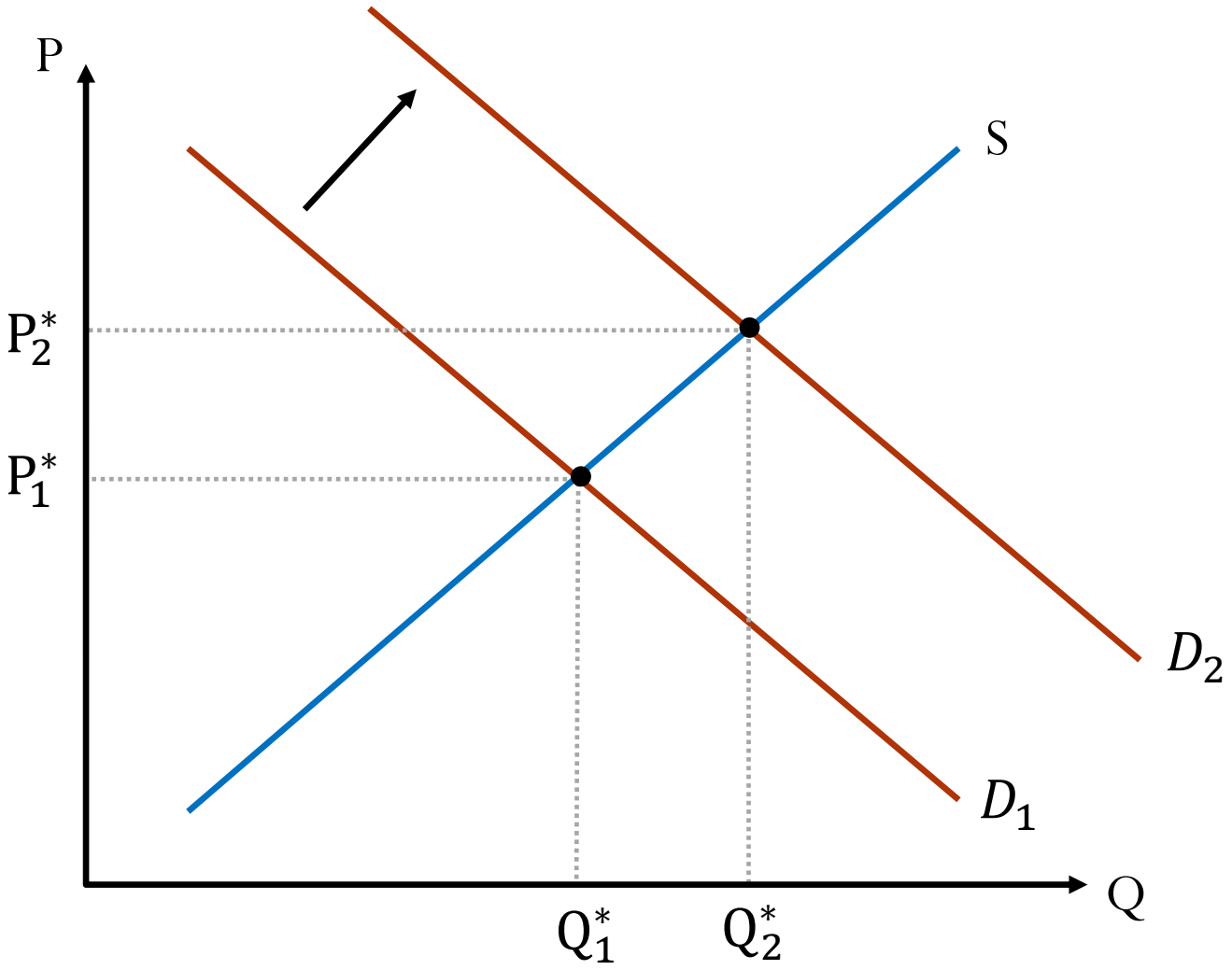
$\Longrightarrow$ Lower prices reduce farmers' incentive to produce.

$\Longrightarrow$ Reduced production would then lead to higher prices…

The final impact of this policy on the market is ambiguous. (Really?)

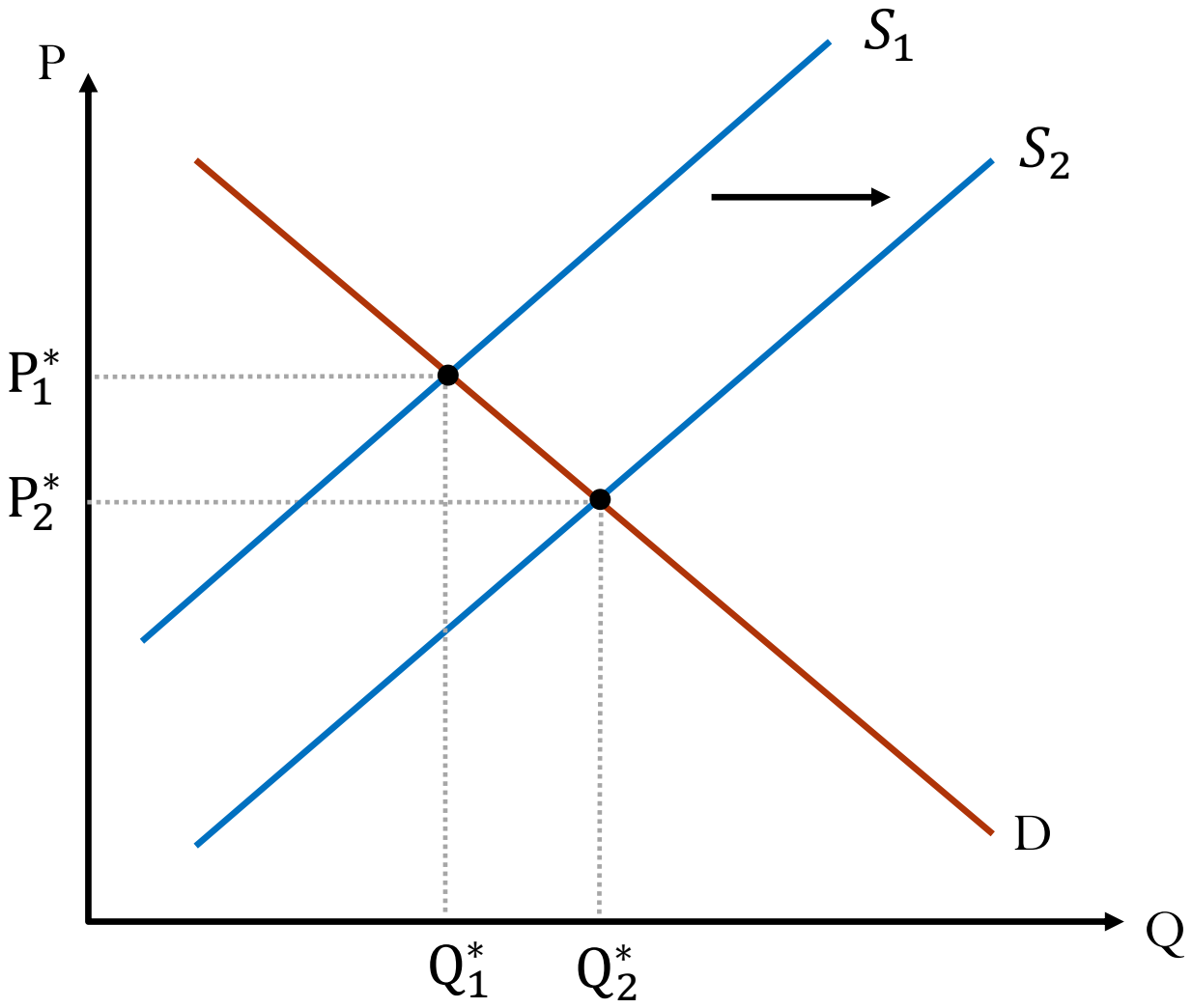
Application I: The Housing Market

- Why do housing prices and sales both rise? Violation of the law of demand?
- First, notice that this must reflect a shift in demand or supply.
- Second, the pattern can only result from a change on the consumer side.



Application II: The Food Market

- Government farm policy leads to increased food production.
- Applying the supply & demand model:
  - The supply curve shifts to the right.
  - At the new equilibrium, price fall, and output rises.
  - The change in market outcomes are actually quite clear!



Application II: The Food Market

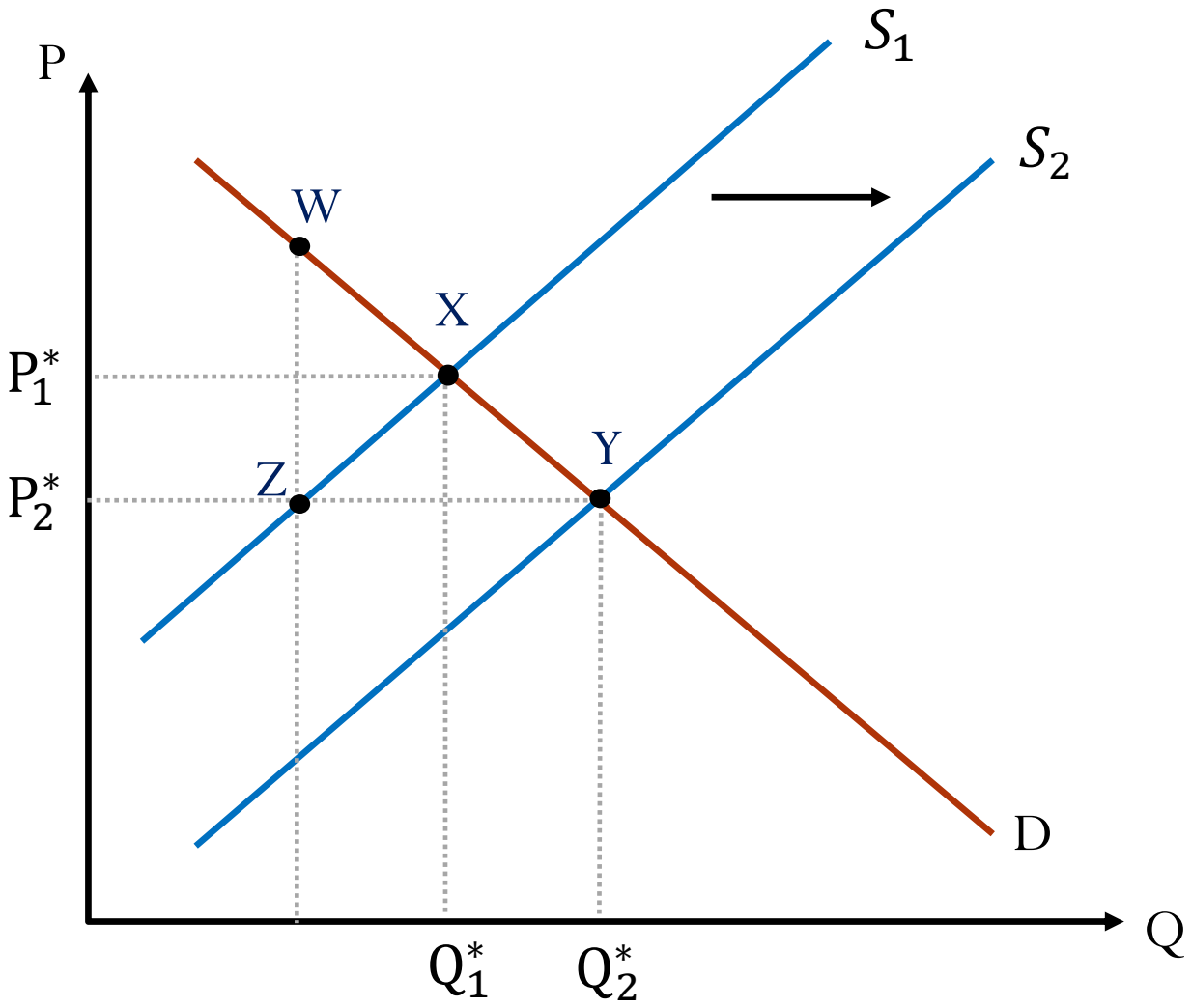
- Government farm policy leads to increased food production.

- Original analysis:

  ⟹ More food supply leads to lower food prices : X → Y ✔

  ⟹ Lower prices reduce farmers' incentive to produce : Y → Z ✘

  ⟹ Reduced production would then lead to higher price: Z → W ✘



Summary

- Demand: definition, changes, factors
- Supply: definition, changes, factors
- Market equilibrium: Supply = Demand
- Comparative statics
- Next, Chapter 4: Consumer and Producer Surplus